CS621 AGENTIC AI FOR INTELLIGENT AUTONOMY · PROJECT PROPOSAL

Career Progress Guide

A consultant-level agent for a job portal that turns a Singaporean's or PR's profile into a costed career plan: realistic next roles, skill gaps, SkillsFuture courses and a direction to follow. Built for career switchers, retrenched jobseekers and anyone upskilling for AI.

Group [#] · [Member names] · [Submission date]

Singaporeans and PRs have generous support to retrain, but must stitch four official sources together to use it

Built for **Career switchers** **Retrenched jobseekers** **Upskillers, incl. for AI** **Mid-career 40+**

$500
SkillsFuture Credit for every citizen 25+

$6,000
Jobseeker Support over 6 months for retrenched citizens and PRs

90%
max fee subsidy for citizens 40+, plus $4,000 Mid-Career Credit

$3,000
monthly cap on the Mid-Career Training Allowance, up to 24 months

What one person has to work out alone, source by source

Where do I stand now?
Skills Framework

→

Which next role is realistic?
Sector guides (PDF)

→

Which skills am I missing?
Skills Framework

→

Which courses close the gap?
Course Directory

→

What do I qualify for and pay?
Funding rule pages

Objective: turn a job-portal profile into a costed career plan the user approves

IN: PROFILE.JSON (GOAL OPTIONAL)

```
{"role": "Accounts Executive", "age": 41, "residency":
"citizen", "status": "retrenched", "goal": "move into data"}
```

→ **The agent**
5 stages, 3 checks with you

→ **OUT: PLAN.JSON**
shown in a Streamlit viewer

The plan holds	Our test asks	Pass mark
Paths: 2–3 roles, e.g. Data Analyst	Is each a real next step from their role?	≥ [x]% of cases
Skill gaps: 13 skills to close	All framework skills found, none extra?	≥ [x]% match
Courses: [1–2] per gap	Do they really teach the missing skills?	≥ [x]% of cases
Direction: start → build → stand out	Every gap placed, simpler levels first?	100% of plans
Cost after funding: $[x], by scheme	Is every dollar and scheme correct?	100% exact
The whole plan, approved by you	Valid JSON? Nothing done without you?	100% valid, 0 actions

Profile schema is our own MyCareersFuture-style mock [assumption: no live portal access].

Profile in, plan out: five LangGraph stages, with the user deciding at three

1 · Table lookup

Profile

Reads profile JSON and goal; maps the job to a framework role

You confirm role and goal reading

2 · Lookup + search

Paths

Uses your goal if given, else nearby roles; LLM ranks 2–3

You choose a path

3 · Table lookup

Gaps

Compares your current role with the path you picked

Agent decides, shows its working

4 · Search (RAG)

Courses

Searches course descriptions for each gap skill

Agent decides, you can swap any

5 · Calculator

Plan

Orders gaps into 3 phases, applies funding rules, writes plan JSON

You approve the plan

Agent does alone

Read, retrieve, compare, calculate

Agent asks you, in Streamlit

Role mapping, chosen path, final plan

Agent never

Edits your profile, enrols, pays or applies

L3 Consultant: the agent owns the analysis, the user owns every judgment call

L1 Operator	L2 Collaborator	L3 Consultant · ours	L4 Approver	L5 Observer
User drives; agent assists on demand	Both plan and execute; user can take over	Agent owns the task, consults on judgment	User sees only blockers and sign-offs	No involvement; logs and off-switch

Why not L2

Co-planning assumes the user already knows skill frameworks and funding rules, the very expertise they lack. Lookups and fee maths are mechanical, and tools do them more reliably than a person.

Why not L4

The right path depends on finances, family and risk appetite, which the agent cannot see. A retrenched user under pressure, given one sign-off, is likely to rubber-stamp.

What would earn L4

Pass rate ≥ [target %] with SD ≤ [x] on every eval set, zero cost errors and zero boundary breaches across k = 3 runs.

Every part of the plan comes from a free, official source

Plan part	Comes from	What's in it	How the agent uses it
Your situation	Portal-style profile (our own mock JSON)	Role, age, residency, job status, salary, goal	Read as the input
Next roles	Skills Framework dataset (sector guide PDFs optional)	Each role's skills and description	Skill overlap finds step-ups; description search finds switches
Skill gaps	Skills Framework dataset + unique skills list	1,910 roles in 39 sectors, ~21 skills each, levels 1–6	Your current role vs. the role you picked
Courses	MySkillsFuture Course Directory	26,099 courses with description, fee, hours	Searches course descriptions by meaning
Cost	SkillsFuture funding rule pages	Credit, subsidies, Jobseeker Support	Coded as calculator rules

What we learned by opening the files

Courses aren't tagged with skills

So the agent must search course descriptions by meaning, not by keyword

The listed subsidised price is one-size-fits-all

So the calculator works out each person's real cost from the rules

Five eval sets, each with ground truth we can compute or check by hand

Eval set	Cases	Ground truth from	Grader	Pass if
E1 Paths & gaps	12 personas × goal / no goal	Skills Framework skill diff; goal must be matched; incl. easy and no-answer cases	Set match (F1)	Valid path; F1 ≥ [x]
E2 Courses	15 gap skills	Two of us label relevant courses (no skill tags exist); report agreement [κ]	Recall@5 vs labels	≥ [x]%
E3 Funding	20 cases	Hand-computed from rule pages: citizen vs PR, age 25 and 40, retrenched or not	Exact match	Exact to the cent
E4 Boundary	10 adversarial profiles	Written first: goals demanding actions, missing fields, hidden instructions	Trajectory assertion	0 breaches in all 3 runs
E5 Plan output	every run	JSON schema; every gap in a phase, lower levels first	Property check	100% valid

Variance

k = 3 runs, mean ± SD; safety must pass all 3 (pass^3)

Two baselines

One LLM call with no tools; a rule-based heuristic with no LLM

Also reported

Cost and latency per task, ask-user rate, failures by tag